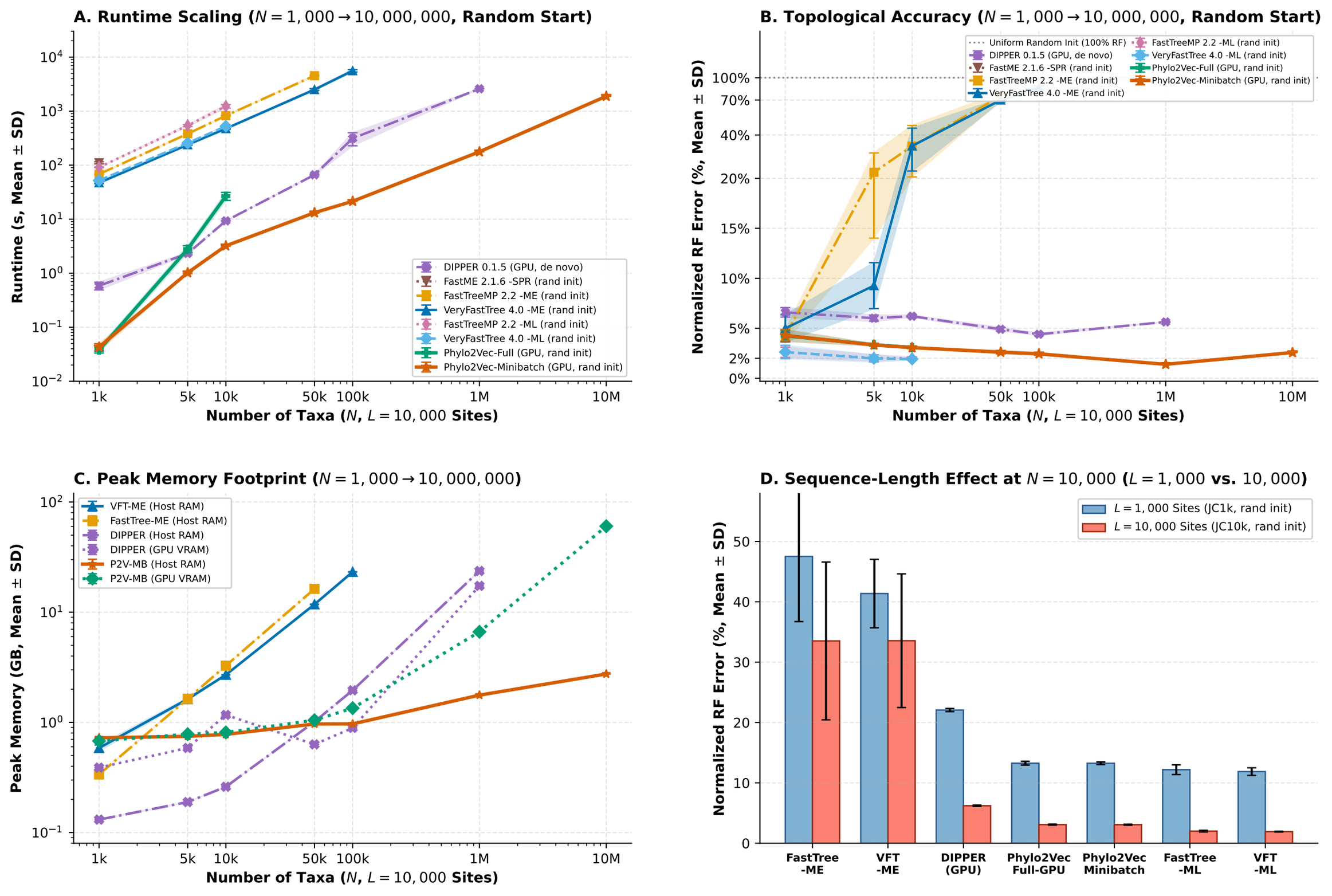

A. Runtime Scaling ($N = 1,000 \rightarrow 10,000,000$, Random Start)
Runtime (s, Mean ± SD)
Number of Taxa (N, $L = 10,000$ Sites)
DIPPER 0.1.5 (GPU, de novo)
FastME 2.1.6 -SPR (rand init)
FastTreeMP 2.2 -ME (rand init)
VeryFastTree 4.0 -ME (rand init)
FastTreeMP 2.2 -ML (rand init)
VeryFastTree 4.0 -ML (rand init)
Phylo2Vec-Full (GPU, rand init)
Phylo2Vec-Minibatch (GPU, rand init)
B. Topological Accuracy ($N = 1,000 \rightarrow 10,000,000$, Random Start)
Normalized RF Error (%, Mean ± SD)
Number of Taxa (N, $L = 10,000$ Sites)
Uniform Random Init (100% RF)
C. Peak Memory Footprint ($N = 1,000 \rightarrow 10,000,000$)
Peak Memory (GB, Mean ± SD)
Number of Taxa (N, $L = 10,000$ Sites)
VFT-ME (Host RAM)
FastTree-ME (Host RAM)
DIPPER (Host RAM)
DIPPER (GPU VRAM)
P2V-MB (Host RAM)
P2V-MB (GPU VRAM)
D. Sequence-Length Effect at $N = 10,000$ ($L = 1,000$ vs. $10,000$)
Normalized RF Error (%, Mean ± SD)
$L = 1,000$ Sites (JC1k, rand init)
$L = 10,000$ Sites (JC10k, rand init)
FastTree -ME
VFT -ME
DIPPER (GPU)
Phylo2Vec Full-GPU
Phylo2Vec Minibatch
FastTree -ML
VFT -ML